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Detection of rice phenology through time series analysis of ground-based spectral index data

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ABSTRACT

Monitoring crop phenology is of great importance for vegetation classification, yield estimation, and irrigation and fertilization management. To test the ability of ground-based remote sensing in detecting major phenological dates of rice, canopy spectra were collected by two portable spectrometers. The Red-edge Chlorophyll Index (CI$_{\text{red edge}}$) and Normalized Difference Vegetation Index (NDVI) time-series derived from ground-based spectrometers were employed to detect the main specific phenological dates. CI$_{\text{red edge}}$ was obtained from ASD FieldSpec Pro spectrometer, while NDVI was from ASD FieldSpec Pro spectrometer (referred to as NDVI$_{\text{ASD}}$) and GreenSeeker RT 100 (referred to as NDVI$_{\text{GS}}$). The phenology detection method consists of two procedures: (i) smoothing the temporal CI$_{\text{red edge}}$ and NDVI data with the double logistic regression function to represent intra-annual vegetation dynamics, (ii) determining the phenological dates through extracting the maximum, minimum and zero-crossing points (FD$_{\text{max}}$, FD$_{\text{min}}$ and FD$_{\text{zero}}$) from the first derivative value of the smoothed NDVI and CI$_{\text{red edge}}$ temporal profiles. A comparison of remote sensing-based estimates with field observations over three growing seasons with different cultivars, planting densities and nitrogen (N) rates showed that CI$_{\text{red edge}}$ can accurately estimate the dates of jointing, middle booting and dough grain. NDVI from both spectrometers can be used to detect the dates of active tillering, middle heading and maturity. Specifically, NDVI$_{\text{GS}}$ yielded better performance than NDVI$_{\text{ASD}}$ for estimating the three phenological dates. Compared with growing season and planting density, rice cultivar and N rate exhibited more significant impact on the accuracy for phenology detection. This work has great potential to provide valuable support for assessing crop growth status and providing precise management strategy. The dates of active tillering, jointing and maturity detected from a combination of CI$_{\text{red edge}}$ and NDVI could be useful for irrigation and fertilization management, and harvest determination, respectively.

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1. Introduction

Phenology information is essential for many applications, such as crop classification (Lloyd 1990; Peña-Barragán et al., 2011; Siachalou et al., 2015; Son et al., 2013), estimation of net primary production (Kimball et al., 2004) and decision-making about water and fertilizer supply (Dingkuhn and Le Gal, 1996). Paddy rice (*Oryza sativa* L.) is one of major food crops in the world, especially in China (Xiao et al., 2002). According to Moldenhauer and Slaton (2001), rice phenology is generally divided into: (i) vegetative phase, including germination, seedling, tillering and jointing stages; (ii) reproductive phase, including booting, heading and flowering stages; (iii) maturation phase, including milk, dough grain and maturity stages. Within these stages, several dates are critical for precision farming management, such as active tillering date (date for field drying), jointing date (date for panicle fertilizer) and maturity date (date for harvesting) (Ling et al., 2007). Irrigation scheduling is critical in the rice growth period, especially in the active tillering stage. Yang et al. (2006) reported that field drying at active tillering stage can control the non-effective tillers and adjust the relationships between soil and water, and those between root and shoot, which are beneficial to rice growth. Panicle fertilizer has great impact on rice grain yield and quality. Suitable application stage of panicle fertilizer could increase the chlorophyll and nitrogen (N) contents of high photosynthetic-rate leaves and thus increase the rice yield (Ding et al., 2003). Harvesting needs to be prompt because late harvesting

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results in reduced milling quality and rice yield (Moldenhauer and Slaton, 2001). All the aforementioned studies demonstrated that the determination of crop phenology is vital for precision management of irrigation and fertilization and determination of harvesting.

Traditionally, crop phenology studies relied on ground-based field visits that were limited by cost, labor and spatial coverage. Consequently, remote sensing (RS) techniques offer considerable benefits for detecting vegetation phenology in the past few decades (Table 1). For example, Tucker et al. (1979) used a hand-held radiometer to monitor corn and soybean growth and development successfully. Gallo and Flesch (1989) utilized the National Oceanographic and Atmospheric Administration's Advanced Very High Resolution Radiometer (NOAA/AVHRR) data for monitoring the seasonal growth of maize at large scale and showed that the date of maximum normalized difference vegetation index (NDVI) agreed well with the silking stage. Meanwhile, several methods have been developed to determine vegetation phenological stages and the most commonly used methods are VI thresholding, inflection point and maximum slope. The VI thresholding method was used by many researchers (Delbart et al., 2006; Fischer 1994; Guo et al., 2016; Markon et al., 1995; Motohka et al., 2010; Nagai et al., 2010). Motohka et al. (2010) defined "GRVI = 0" as a site-independent single threshold for detecting the early phase of leaf green-up and the middle phase of autumn coloring in four different vegetation types. With the inflection point method, the start of the growing season can be identified when the first derivative (FD) value of the time-series curve changes from negative to positive, while the end of the growth can be determined when it changes from positive to negative (Moulin et al., 2010; Sakamoto et al., 2005; Soudani et al., 2008; Zhang et al., 2003). With the maximum slope method, the growth stages are determined by the magnitude of variation in the VI time series (Wang et al., 2014a; Yu et al., 2003).

To date, most crop phenology studies relied on satellite RS data due to the widespread availability of high temporal resolution time series imagery from such instruments as the Moderate Resolution Imaging Spectroradiometer (MODIS) and the NOAA/AVHRR. Although these satellite imagery can cover large areas, they are affected by many factors, such as atmospheric disturbances, solar radiation effects and cloud cover duration (Ricotta and Avena 2000; White et al., 2005). Therefore, the estimation error of phenology could be as high as seven days or even more (Sakamoto et al., 2005; Sun et al., 2009). Coarse resolution satellite imagery is not suitable for phenology detection in major rice growing regions such as south China because the rice fields in this region are relatively small, irregular and fragmented by well-developed roads and dense water networks (Wang et al., 2015). To the best of our knowledge, there was no report about phenology detection using airborne or unmanned aerial vehicle (UAV) imagery primarily due to the logistics complexity and resource limitation with frequent revisits over the growing season. Hence, ground-based RS has great advantages on phenology detection in south China due to its flexibility of spectral, spatial and temporal resolutions.

Recently, time series VIs from ground-based RS platforms have been applied to phenology studies. NDVI was commonly used in phenological studies and had good performance in phenology detection as shown in a summary of relevant references in Table 1. For instance, Wang et al. (2014a) used NDVI from a FieldSpec3 spectroradiometer and its slope curves to monitor rice development and demonstrated that they could be used as cultivar-independent phenological indicators. While handheld sensors with more bands become available in recent years for proximal monitoring of crops, more spectral indices could be obtained to detect crop vigor using red edge bands instead of the red band in NDVI. As one of the red edge indices, red-edge chlorophyll index ($CI_{red\ edge}$) was originally proposed for chlorophyll estimation and later be found to be a better indicator for leaf area index (LAI) and biomass than NDVI (Gitelson et al., 2003a).

For rice phenology detection, several stages (e.g. transplanting, tillering, heading and harvesting stages) were determined in previous studies (Motohka et al., 2010; Sakamoto et al., 2005; Wang et al., 2014a). However, other dates (e.g. the dates of active tillering and jointing) that are considered to be more important for irrigation scheduling and fertilizer management, have received little attention. To fill this gap in the previous phenological studies, this study attempts to detect all those main phenological dates of paddy rice with $CI_{red\ edge}$ and NDVI. Our research objectives are: (1) evaluating the performance of a chlorophyll index ($CI_{red\ edge}$) for rice phenology detection in comparison to the widely used NDVI; (2) determining rice critical phenological dates with the maximum, zero-crossing and minimum points from the first derivative (FD_{max}, FD_{min} and FD_{zero}) of VI time series. We expected that our work would provide useful guidance for water and fertilizer management and harvesting with portable spectrometer devices.

2. Materials and methods

2.1. Experiment design

Three field experiments were designed for this study, involving different rice cultivars, planting densities and N rates. All the experiments were conducted in the experimental station of National Engineering and Technology Center for Information Agriculture (NETCIA), which was located in Rugao city, Jiangsu province, China (120°45′ E, 32°16′ N). In 2013, one japonica rice cultivar Wuxiangjing14 (V1) and one indica rice cultivar Shanyou63 (V2) were seeded at DOY 138 and transplanted into the paddy field at DOY 172. Four N rates (0 (N0), 150 (N1), 300 (N2) and 375 (N3) kg N ha^{-1} as urea) were applied with 40% at preplanting, 10% at tillering, 30% at jointing and 20% at booting. Treatments with N rates of 0 and 375 kg N ha^{-1} were transplanted with a planting density of 220,000 plants ha^{-1} (D1), while treatments with N rates of 150 and 300 kg N ha^{-1} were transplanted with two planting densities of 220,000 plants ha^{-1} (D1) and 130,000 plants ha^{-1} (D2). The plot size was 30 m^2 with 6 m length and 5 m width. For all treatments, monocalcium phosphate and potassium chloride were applied prior to transplanting at 135 kg P_2O_5 ha^{-1} and 220 kg K_2O ha^{-1}. The experiment was in a randomized complete block design with three replications for each treatment. Other management followed local standard practices in rice production.

Experiment 2 and 3 were similar to the experiment 1. The cultivars were japonica rice cultivar Wuyunjing24 (V1) and indica rice cultivar Yliangyou1 (V2), which were seeded at DOY 135 and transplanted at DOY 167 in 2014. In 2015 growing season, rice was seeded at DOY 136 and transplanted at DOY 166. Four N rates (0 (N0), 100 (N1), 200 (N2) and 300 (N3) kg N ha^{-1} as urea) were applied as same as the experiment 1. Rice gains were harvested at DOY 286, 292 and 299 in 2013, 2014 and 2015, respectively.

2.2. Data collection

Two sensors were used to measure canopy reflectance spectra in this study: (1) ASD FieldSpec Pro spectrometer (Analytical Spectral Devices, Boulder, CO, USA) with a spectral range of 350–2500 nm and a 25° view angle, abbreviated as ASD; (2) GreenSeeker RT 100 (NTech Industries, Ukiah, CA, USA), abbreviated as GS. GS was selected as the reference active instrument due to its most wide use for in situ crop monitoring and the direct availability of NDVI. Measurements of each treatment were taken on the same day with irregular observation intervals varying from 3 to 10 days, with an average of 5 days. During sunny daytime, the measurements were

Table 1
Summary of phenology detection with remote sensing. The references are indexed by the year of publication and summarized with the species studied, the method used in the paper, the phenological stages and the source of remote sensing data. The studies on rice crop were marked with *.

Reference	Species	Method	Phenological stages	Data source
Sakamoto et al. (2005)	paddy rice*	EVI, inflexion point method	Planting date, heading date, harvesting date, and growing period	MODIS
Sun et al. (2009)	Paddy rice*	EVI, low pass Fourier and wavelet filtering methods	Transplanting, beginning of tillering, heading, and maturation stages	MODIS
Boschetti et al. (2009)	Rice crop*	TIMESAT and GDD	Emergence, heading and maturity	MODIS
Wang et al. (2014a)	Paddy rice*	NDVI, original curve and its slope	Flooding and transplanting, tillering, panicle development, heading and flowering, maturity, harvest stages	FieldSpec3 spectroradiometer
Ishikawa et al. (2015)	Paddy rice*	Second derivative spectra of 400–1100 nm	Active tillering stage, maximum tiller number stage, heading and mature grain stage	A handheld spectrometer (KE-1)
Zhang et al. (2003)	Global vegetation	EVI, piecewise logistic functions	Greenup, maturity, senescence and dormancy dates	MODIS
Vina et al. (2004)	maize	NDVI and $\text{VARI}_{\text{Green}}$, Scaled first derivative	Tassel emergence and start of senescence	Ocean optics spectroradiometer
Delbart et al. (2006)	boreal ecosystems	NDWI-VGT method and NOAA-NDVI pixel specific threshold	Onset of greening	SPOT-VGT and NOAA AVHRR
Soudani et al. (2008)	deciduous forest	NDVI, inflexion point method	Onset of green-up	MODIS
Sakamoto et al. (2010)	maize and soybean	WDRVI, Two-step filtering	Maize: early vegetative stage, silking stage, dent stage and maturity; soybean: early vegetative stage, beginning seed, full seed and beginning maturity	MODIS
Motohka et al. (2010)	deciduous broadleaf forest, deciduous coniferous forest, grassland and rice paddy field	GRVI and NDVI, threshold method	Early phase of leaf green-up and middle phase of autumn coloring	A hemi-spherical spectro-radiometer and a fish-eye camera
Nagai et al. (2010)	a deciduous broad-leaved forest	NDVI, threshold method	Timing of leaf expansion and defoliation	An automatic digital fish-eye camera system
Knox et al. (2013)	Grasses	PhIX, statistics method	Seed, adult, flower and dormancy	GER 3700 spectroradiometer
Pan et al. (2015)	winter wheat and summer corn	NDVI, TIMESAT program	Start and end of the season	Chinese HJ–1 A/B satellites
Guo et al. (2016)	winter wheat	NDVI, relative threshold method and dynamic threshold method	Start of spring season	NOAA AVHRR

NOAA = National oceanic and atmospheric administration, AVHRR = Advanced very high resolution radiometer, MODIS = Moderate resolution imaging spectroradiometer, NDVI = Normalized difference vegetation index, EVI = Enhanced vegetation index, VARI = Visible atmospherically resistant index, GRVI = Green-red vegetation index.

performed at 1.0 m above the canopy between 11:00 a.m. and 1:00 p.m. local time. For ASD, there were 3 fixed observation points in each plot, and each point was recorded five times. The averaged spectrum was calculated to represent the plot value. Optimization and white panel reflectance measurements were performed every 10 min. For GS, there were 3 fixed observation trips in each plot and the trigger was held for 3 m distance. The averaged spectrum of these three strips was calculated to represent the plot value. Frequent field observations were carried out and the exact date of main phenology stages (green returning, tillering, jointing, booting, heading, milk filling and maturity) were recorded.

2.3. Spectral indices

In this study, the NDVI (Rouse et al., 1974) and the $\text{CI}_{\text{red edge}}$ (Gitelson et al., 2003b) were employed to estimate the rice phenology. They were calculated as:

$$\text{CI}_{\text{red edge}} = (\rho_{\text{NIR}}/\rho_{\text{Red Edge}}) - 1 \tag{1}$$

$$\text{NDVI} = (\rho_{\text{NIR}} - \rho_{\text{Red}})/(\rho_{\text{NIR}} + \rho_{\text{Red}}) \tag{2}$$

where ρ_{NIR}, $\rho_{\text{Red Edge}}$ and ρ_{Red} are the reflectance in the wavelength of 800 nm (770 nm in NDVI_{GS}), 720 nm and 680 nm (660 nm in NDVI_{GS}), respectively.

2.4. First-derivative analysis

To obtain high quality time series of vegetation indices, a double logistic regression function (Beck et al., 2006) was applied to smooth the original data. This smooth method was suitable for vegetation with one growth cycle (Qiu et al., 2016). It can hold outliers effectively and can be used to estimate parameters related to phenological events (Beck et al., 2006). To identify the dates of key phenological transitions, first derivative (FD) analysis was applied to the temporal profiles of NDVI and $\text{CI}_{\text{red edge}}$. For that, the FD of the VI with respect to day of year (DOY) was calculated as:

$$\text{FD}_{\text{VI}} = (\text{VI}_{(j+1)} - \text{VI}_{(j)})/\Delta_{\text{DOY}} \tag{3}$$

where FD_{VI} is the FD value of VI (NDVI or $\text{CI}_{\text{red edge}}$) between the jth and j + 1th DOY, $\text{VI}_{(j)}$ is the VI value at the jth DOY, $\text{VI}_{(j+1)}$ is the VI value at the j + 1th DOY, and ΔDOY equals 1 as the interval.

2.5. Threshold determination for detecting phenological dates

2.5.1. Phenological dates estimated with NDVI

Previous studies have reported that the maximum NDVI appears at the heading stage because of a high level of LAI and biomass at this stage (Motohka et al., 2010; Wang et al., 2014a). As the heading and flowering stage lasts about 9 days for the entire field, we used the date of zero-crossing point on the FD (FD_{zero}) of NDVI to determine the middle heading date. Piao et al. (2006) and Studer et al. (2007) reported that the day with the maximum increase rate of NDVI can be determined as the onset of leaf expansion, and the day with the maximum decrease rate of NDVI as the offset of defoliation. Similarly, rice plants grow very fast at the tillering stage, and the highest speed might be at the active tillering stage. Therefore, the date of the maximum point on the FD (FD_{max}) of NDVI was used to estimate the active tillering date. Additionally, there might be a rapid decrease of NDVI at the end of maturation stage, which is due to the leaf senescence. Hence, we used the date of the minimum point on the FD (FD_{min}) of NDVI to detect the maturity date.

2.5.2. Phenological dates estimated with $\mathit{CI}_{\mathit{red\ edge}}$

Gitelson et al. (2003a) stated that $\text{CI}_{\text{red edge}}$ can accurately estimate LAI and green leaf biomass superior to the traditional NDVI, due to its insensitivity to moderate-to-high LAI and biomass. Furthermore, $\text{CI}_{\text{red edg}}$ has a good capability in estimating chlorophyll content, even in moderate or high chlorophyll content level (Gitelson et al., 2005; Gitelson et al., 2003b). In general, these agronomic variables (e.g. LAI, biomass, chlorophyll content) increase fast over the vegetative stage and reach their peak at the booting stage in rice. We assumed that the maximum increase of $\text{CI}_{\text{red edge}}$ is at the jointing stage and the peak of $\text{CI}_{\text{red edge}}$ is at the booting stage. Therefore, we used the date of the maximum point and zero-crossing point on the FD (FD_{max} and FD_{zero}) of $\text{CI}_{\text{red edge}}$ to estimate the dates of jointing and middle booting, respectively. After the heading stage, rice grains begin filling with the carbohydrate, and the chlorophyll content decreases dramatically as leaves senesce. The maximum decrease is approximately on the date of dough grain. Therefore, the date of minimum point on the FD (FD_{min}) of $\text{CI}_{\text{red edge}}$ was used to determine the dough grain date.

3. Results

3.1. NDVI based estimation of phenological dates

3.1.1. Overall trend of NDVI

Time series of NDVI and its FD value under different conditions are shown in Fig. 1. The curve of the treatment with Wuyunjing24 under 200 kg ha^{-1} N rate and a planting density of 220,000 plants ha^{-1} (V1N2D1) in 2014 (black symbols in Fig. 1a) was taken as an example. NDVI increased rapidly from DOY 175 (initial tillering) to 211 (initial jointing) and its maximum increase rate (FD_{max}) was at DOY 192 (active tillering). NDVI reached the maximum around 0.93 at DOY 230 (middle booting) and then remained stable between DOY 240 (initial heading) and 249 (initial filling). After that, NDVI began to decline slowly, and its FD value changed from positive to negative. The decrease trend appeared from DOY 250 (initial filling) to 288 (maturity) and decreased more significantly since DOY 268 (dough grain). At DOY 289 (maturity), the FD of NDVI reached its minimum.

3.1.2. Effects of cultivar and cultivation factors on NDVI

Year of growing season and planting density had no significant influence on the shape of NDVI time series (Fig. 1a and b). However, compared with the temporal profiles of NDVI in japonica rice (V1), a much sharper increase rate was observed in indica rice (V2) (Fig. 1c). On the other hand, indica rice had a longer plateau than japonica rice. For different N rates, the temporal profile of NDVI without N application had a slower increase rate and lower plateau than other N treatments (Fig. 1d). Fig. 2 shows the difference of NDVI time series from different sensors. Apparently, NDVI_{ASD} had an earlier and higher maximum of the FD value than NDVI_{GS}. Moreover, the plateau in NDVI_{ASD} time series was higher and longer than that in NDVI_{GS} time series, but both of them decreased to the minimum value nearly at the same time.

Table 2
Root mean square error (RMSE) in day for the phenological dates estimated by NDVI against the field observations. AT, MH and MT represent the active tillering, middle heading and maturity, respectively.

	Treatment	AT date	MH date	MT date
NDVI_{ASD}	All	5.8	11.0	4.8
	2013	5.6	12.5	5.2
	2014	6.1	10.9	4.4
	2015	5.6	9.1	4.8
	D1	6.5	11.2	4.6
	D2	6.4	11.0	5.5
	V1	3.5	11.1	6.4
	V2	6.8	10.8	2.4
	N0	3.4	7.9	5.3
	N1	6.4	12.2	4.4
	N2	6.6	10.1	4.8
	N3	4.6	12.8	3.9
NDVI_{GS}	All	4.9	11.5	3.2
	2013	–	–	–
	2014	4.8	13.4	4.1
	2015	5.0	9.3	1.8
	D1	6.0	12.8	2.2
	D2	4.7	7.4	3.6
	V1	3.0	7.3	3.9
	V2	6.2	14.6	1.1
	N0	3.1	9.2	3.4
	N1	6.6	12.3	1.6
	N2	5.4	13.3	2.7
	N3	4.2	16.6	2.9

3.1.3. Phenology detection based on $\mathit{NDVI}_{\mathit{ASD}}$ and $\mathit{NDVI}_{\mathit{GS}}$

The results based on NDVI estimation for the three specific phenological dates of rice are presented in Table 2. A comparison between the field observations and the RS-based estimates is shown in Figs. 3 and 4. NDVI from the two sensors performed best in estimating maturity date with all estimation error within ±10 days (Figs. 3 c and 4 c). Compared with NDVI_{ASD}, NDVI_{GS} yielded a higher accuracy in maturity date estimation with an RMSE of 3.2 days (Table 2). Similarly, the performance of NDVI_{ASD} was not so good as NDVI_{GS} in estimating the active tillering date, with RMSEs from 5.8 days to 4.9 days (Table 2). Both NDVI_{GS} and NDVI_{ASD} yielded low estimation accuracy in detecting middle heading date, with RMSEs over 11 days for the whole database (Table 2). There was no significant difference between estimates in growing seasons and planting densities for estimating the active tillering and maturity dates (the difference in RMSE ranged from 0.1 to 2.4 days; Table 2). For different rice cultivars, NDVI performed better in japonica (V1) rice for active tillering date estimation and worse for maturity date estimation than NDVI in indica (V1) rice. There was no consistent trend (earlier or later) in the differences between the estimated and the observed dates for these three specific phenological dates under different N rates (Table 2). The estimation accuracy of middle heading date varied greatly under different conditions, except for the NDVI_{ASD} in different planting densities.

3.2. $\mathit{CI}_{\mathit{red\ edge}}$ based estimation of phenological dates

3.2.1. Overall trend of $\mathit{CI}_{\mathit{red\ edge}}$

Time series of $\text{CI}_{\text{red edge}}$ and its FD value under different conditions are shown in Fig. 5. The curve of the treatment with Wuyunjing14 under 200 kg ha^{-1} N rate and a planting density of 220,000 plants ha^{-1} (V1N2D1) in 2014 (black symbols in Fig. 5a) was taken as an example. The value of $\text{CI}_{\text{red edge}}$ increased from DOY 175 (initial tillering) to DOY 232 (middle booting) and a maximum increase point (FD_{max}) appeared at DOY 212 (jointing). At DOY 234 (middle booting), $\text{CI}_{\text{red edge}}$ reached its peak, and the FD of

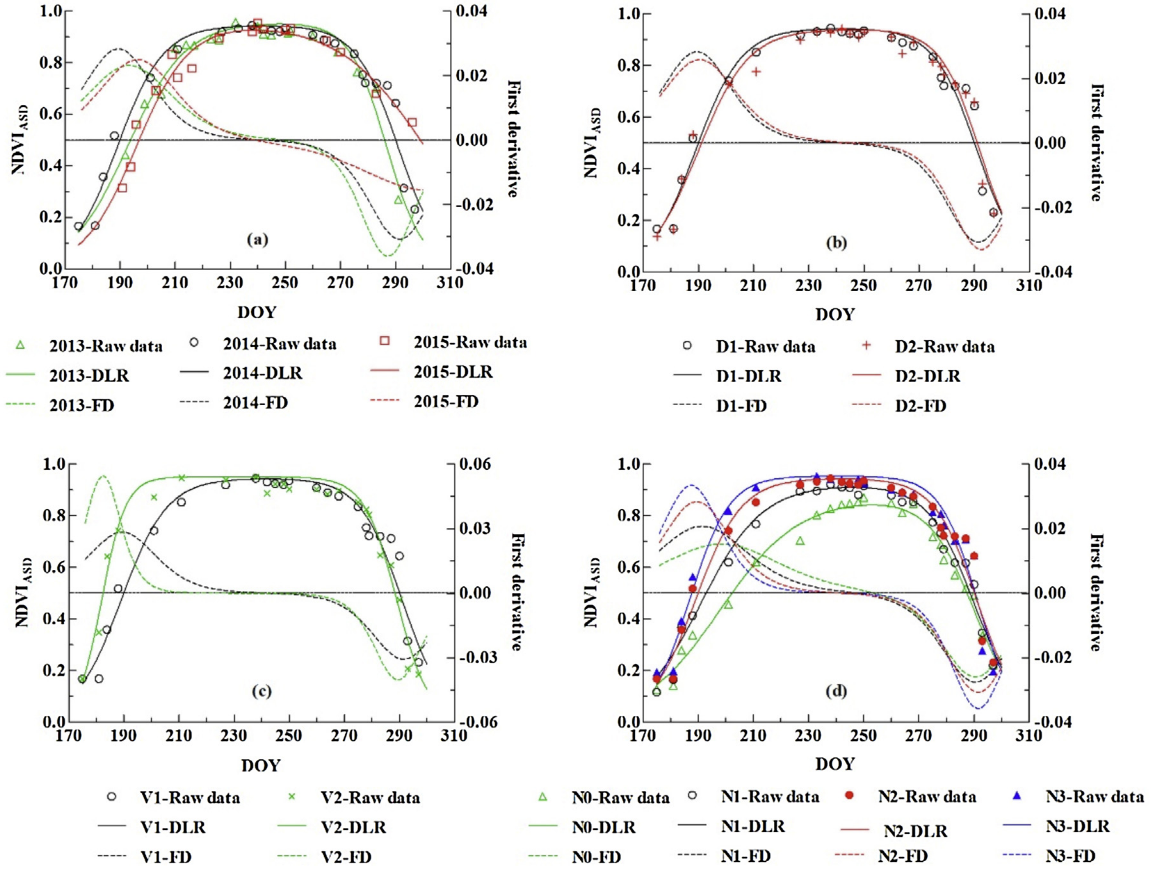

Fig. 1. Temporal profiles of $\mathrm{NDVI}_{\mathrm{ASD}}$ and its first derivative under different conditions of growing season (a), planting density (b), rice cultivar (c) and nitrogen rate (d). The circles indicate the original NDVI data. The solid lines are fitted by double logistic regression. The dashed lines are the first derivatives of the fitted NDVI in solid lines. DLR = double logistic regression, FD = first derivative. V1 and V2 represent Wuyunjing24 and Yliangyou1; D1 and D2 represent 220,000 plants ha^{-1} and 130,000 plants ha^{-1}; N0, N1, N2, and N3 represent 0, 100, 200 and 300 kg N ha^{-1} as urea, respectively.

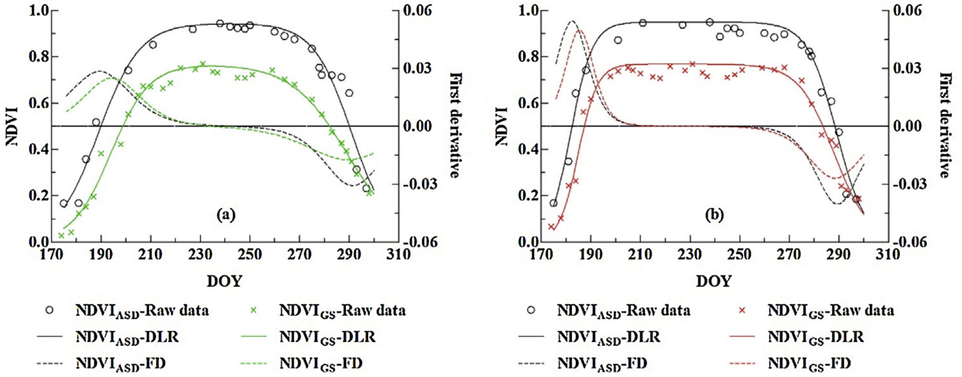

Fig. 2. Temporal profiles of $\mathrm{NDVI}_{\mathrm{ASD}}$, $\mathrm{NDVI}_{\mathrm{GS}}$ and their first derivatives for Wuyunjing24 (a) and Yliangyou1 (b). The circles indicate the original NDVI data. The solid lines are lines fitted by double logistic regression. The dashed lines are the first derivative value of NDVI. DLR = double logistic regression, FD = first derivative.

$\mathrm{CI}_{\mathrm{red\ edge}}$ changed from positive to negative. After the peak, $\mathrm{CI}_{\mathrm{red\ edge}}$ decreased gradually and its maximum decrease ($\mathrm{FD}_{\mathrm{min}}$) was at DOY 265 (dough grain).

3.2.2. *Effects of cultivar and cultivation factors on $\mathrm{CI}_{\mathrm{red\ edge}}$*

Fig. 5a and b show the year of growing season and planting density had no significant influence on the profiles of $\mathrm{CI}_{\mathrm{red\ edge}}$. Compared with the temporal profiles of $\mathrm{CI}_{\mathrm{red\ edge}}$ in japonica rice (V1), the $\mathrm{FD}_{\mathrm{max}}$ value appeared earlier and higher in indica rice (V2) (Fig. 5c). Moreover, the peak of $\mathrm{CI}_{\mathrm{red\ edge}}$ reached its peak earlier and the maximum decrease point ($\mathrm{FD}_{\mathrm{min}}$) appeared earlier in indica rice (Fig. 5c). For different N rates, it was obvious that the temporal profile of $\mathrm{CI}_{\mathrm{red\ edge}}$ under N0 treatment had a slower increase and lower maximum than other treatments (Fig. 5d).

3.2.3. *Phenology detection with $\mathrm{CI}_{\mathrm{red\ edge}}$*

The summary of the estimation accuracy for the three specific phenological dates estimated by $\mathrm{CI}_{\mathrm{red\ edge}}$ is shown in Table 3 and a comparison of those estimated dates with the field observations is shown in Fig. 6. The phenological date with the highest estimation accuracy was the jointing date with all estimation errors within ±10 days (Fig. 6a). In addition, the performance of $\mathrm{CI}_{\mathrm{red\ edge}}$

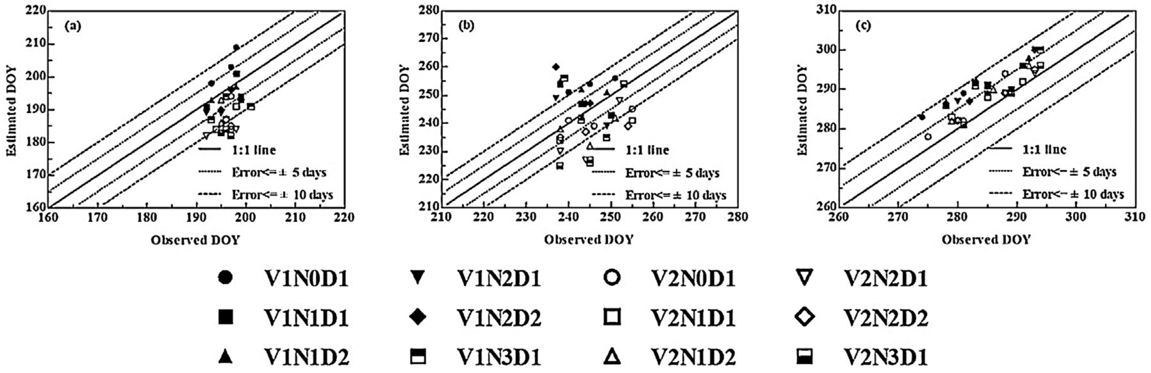

Fig. 3. Comparison between field observed and $NDVI_{ASD}$-derived phenological dates for active tillering (a), middle heading (b) and maturity (c). DOY = day of year.

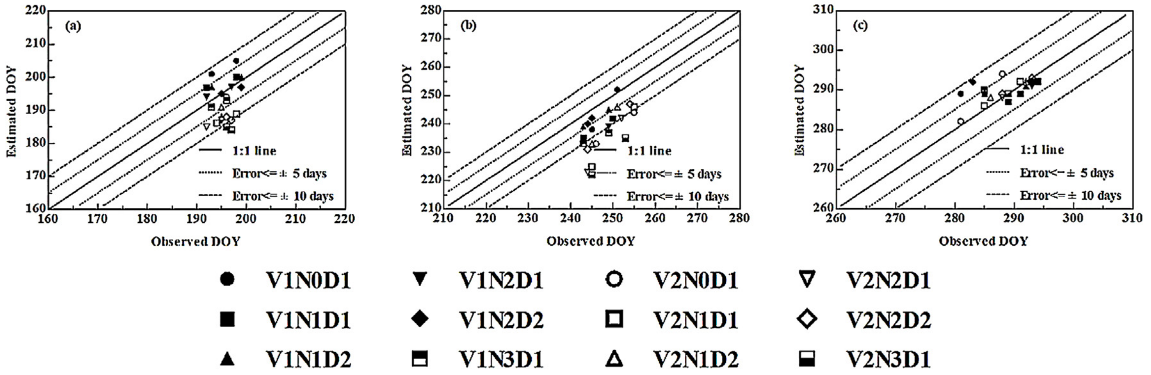

Fig. 4. Comparison between field observed and $NDVI_{GS}$-derived phenological dates for active tillering (a), middle heading (b) and maturity (c). DOY = day of year.

Table 3
Root mean square error (RMSE) in day for the phenological dates estimated by $CI_{red\ edge}$ against the field observations. JT, MB and DG represent jointing, middle booting and dough grain, respectively.

Treatment	JT date	MB date	DG date
All	3.6	6.2	6.6
2013	3.6	6.4	7.4
2014	3.5	5.2	5.0
2015	3.6	7.0	7.2
D1	4.2	5.6	6.0
D2	3.4	5.6	6.9
V1	2.8	7.7	7.9
V2	4.2	4.2	5.1
N0	2.3	9.3	8.3
N1	3.8	5.3	6.0
N2	4.6	5.9	6.1
N3	3.5	4.5	5.3

was stable under different conditions (RMSE ranged from 2.3 to 4.6 days (Table 3)). $CI_{red\ edge}$ had moderate performance in estimating dough grain date with an RMSE of 6.6 days, which was comparable to the estimation of middle booting date with an RMSE of 6.2 days (Table 3). There was no significant difference between estimates in growing seasons and planting densities for detecting the three specific phenological dates (the difference in RMSE ranged from 0 to 2.4 days; Table 3). For different rice cultivars, the estimates of japonica (V1) were closer to the 1:1 line than those of indica (V2) for jointing date estimation (Fig. 6a). $CI_{red\ edge}$ yielded better performance in indica rice than that in japonica rice for middle booting date and dough grain date estimation (Figs. 6b and c). There was no consistent trend (earlier or later) in the differences between the estimated and the observed dates for these three specific phenological dates under different N rates (Table 3).

4. Discussion

4.1. Advantages of $CI_{red\ edge}$ in phenology detection

N is the most important element which impacts yield and quality in crops (Cassman et al., 2002; Shanahan et al., 2008), and panicle fertilizer applied at jointing stage has a great impact on yield formation (Ding et al., 2003). However, this stage was not addressed in previous phenological studies. Our results suggest that $CI_{red\ edge}$ is a good indicator for phenology detection. The FD_{max}, FD_{zero} and FD_{min} points of $CI_{red\ edge}$ temporal profile correspond well to the jointing, middle booting and dough grain dates, respectively. Rice plants begin to grow one week after transplanting, then LAI and biomass increase fast at the tillering stage and continue to increase at the jointing stage and their peaks appear at booting stage. The maximum leaf chlorophyll content also occurs at the booting stage (Wang et al., 2014b). As $CI_{red\ edge}$ does not saturate at high LAI, biomass and chlorophyll content (Gitelson et al., 2003a; Gitelson et al., 2005), $CI_{red\ edge}$ increases with the plant growth. Therefore, the maximum increase of $CI_{red\ edge}$ (FD_{max}) can be used to detect the jointing date. After booting stage, chlorophyll content decreases with the coming heading and flowering. Hence, the FD_{zero} point of $CI_{red\ edge}$ agrees well with the middle booting date. The FD_{min} point of $CI_{red\ edge}$ corresponds to the dough grain date due to the dramatic decrease of chlorophyll content at the grain filling and ripening stages.

4.2. Effects of cultivar and cultivation factors on the estimation accuracy

Both NDVI and $CI_{red\ edge}$ had stable performance in phenology estimation (except for the middle heading date) under different

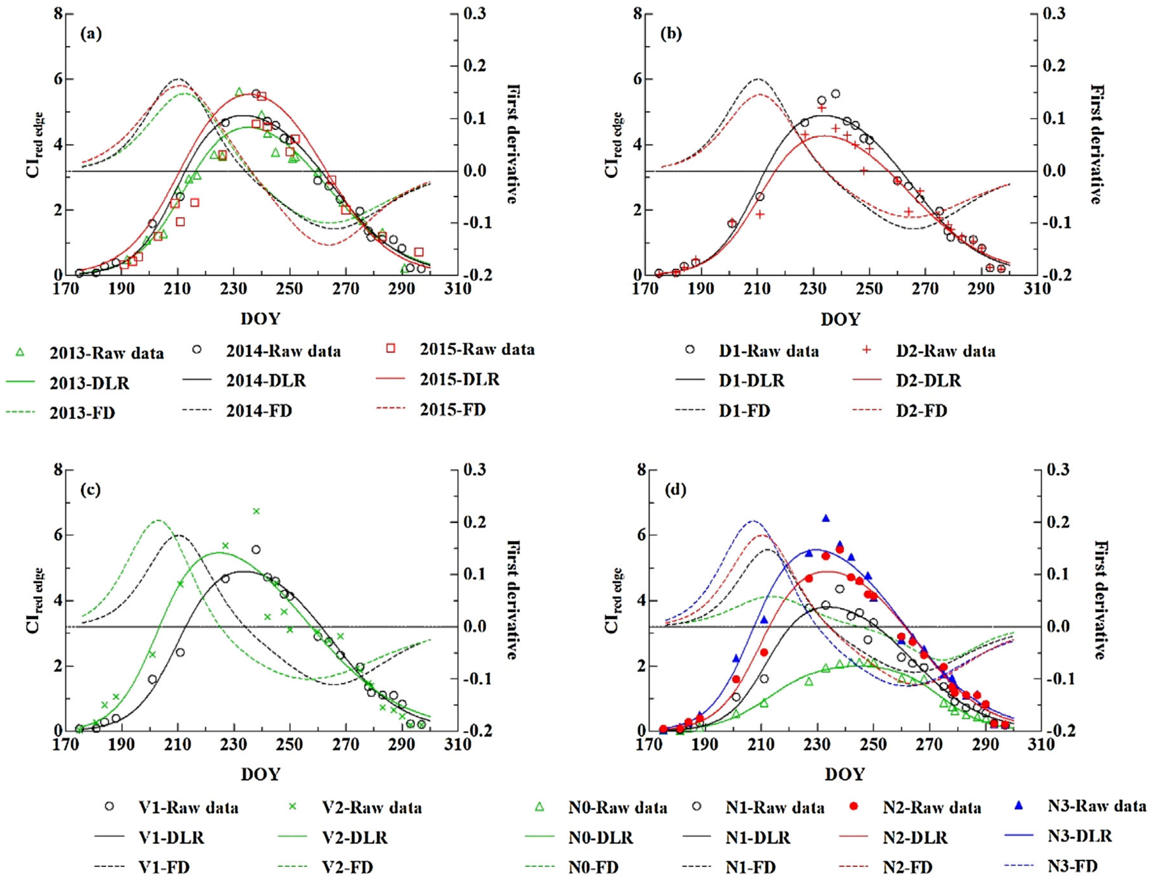

Fig. 5. Temporal profiles of $CI_{red\ edge}$ and its first derivative value under different conditions of growing season (a), planting density (b), rice cultivar (c) and nitrogen rate (d). The circles indicate the original $CI_{red\ edge}$ data. The solid lines are lines fitted by double logistic regression. The dashed lines are the first derivative value of $CI_{red\ edge}$. DLR = double logistic regression, FD = the first derivative value. V1 and V2 represent Wuyunjing24 and Yliangyou1; D1 and D2 represent 220,000 plants ha^{-1} and 130,000 plants ha^{-1}; N0, N1, N2, and N3 represent 0, 100, 200 and 300 kg N ha^{-1} as urea, respectively.

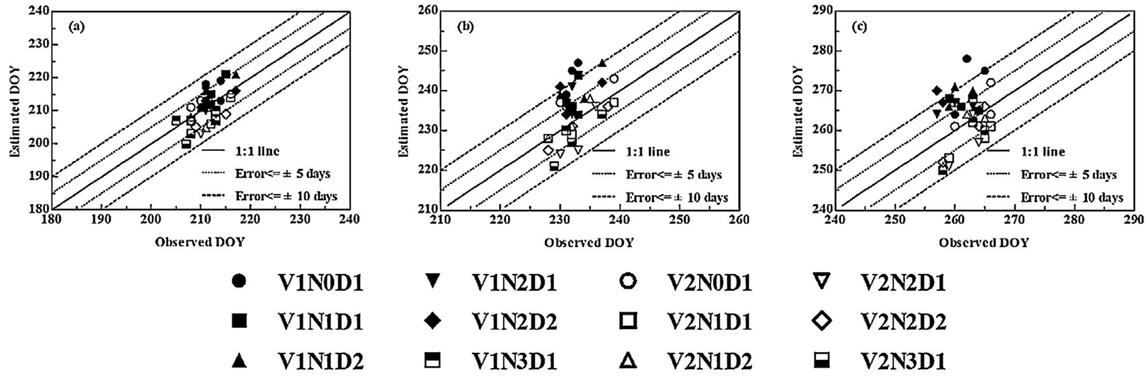

Fig. 6. Comparison between field observed and $CI_{red\ edge}$-derived phenological dates for jointing (a), middle booting (b) and dough grain (c). DOY = day of year.

years and planting densities with a difference in RMSE less than 2.4 days and less than 2.3 days, respectively (Tables 2 and 3). The reasons might be that planting density had no significant influence on the chlorophyll content (data not shown). NDVIs under different planting densities were close though slightly higher LAI and biomass were observed in dense planting density than those in spares.

Compared with the effects of growing season and planting density, rice cultivar has a greater impact on the estimation accuracy (Tables 2 and 3), which might be explained by the following three reasons: (1) indica rice has a stronger ability of producing tillers; (2) LAI and biomass reach their peaks earlier in indica rice; (3) dropping temperature in the grain filling stage makes indica rice mature earlier. These reasons are supported by Zhang et al. (2013). Furthermore, the estimation accuracy varies greatly under different N rates (Tables 2 and 3), because N stress makes rice grow slowly in the vegetation stage and senesce fast in the reproductive and ripening stages. Additionally, rice plants with higher N content not only have higher chlorophyll content but also thicker, flatter and more turgid leaves, which causes higher NDVI and $CI_{red\ edge}$ values (Lee et al., 2011).

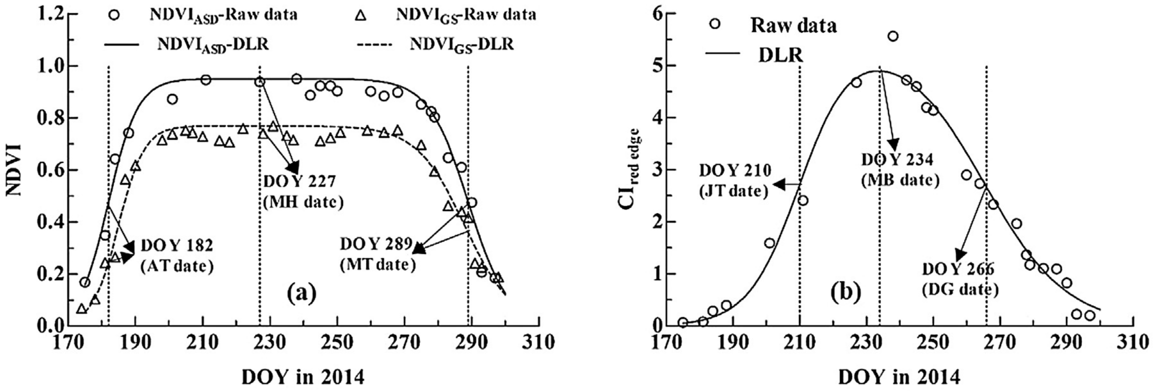

Fig. 7. NDVI (a) and $CI_{red\ edge}$ (b) derived phenological dates indicated by arrows and marked on the temporal profiles smoothed with the double logistic regression.

The most difficult date to estimate is middle heading date due to the severe saturation of NDVI (Gitelson et al., 2002), which makes the estimates range wider. NDVI has a moderate performance in active tillering date estimation, which can be explained with the effects of soil background and variations in canopy structure on canopy reflectance signals at early growth stages (Daughtry et al., 2000; Haboudane et al., 2002). Reed et al. (1994) found that the end-of-greenness metric was related to the most rapid decline in NDVI, and the similar result was found in this study that the dramatic decrease of NDVI agreed well with the maturity date. Therefore, effects of growth stage, crop type and the variation of chlorophyll content should be taken into account when applying RS on phenology detection.

4.3. *Comparison between $NDVI_{ASD}$ and $NDVI_{GS}$*

In general, NDVI from the active spectral senor (GreenSeeker) had better performance than that from the passive one (ASD) in phenological dates detection (Table 2), which can be explained with the following two reasons. First, ASD is a passive sensor, which needs to be calibrated with white panel before collecting the canopy reflectance. Solar radiation might change during the measurement and error could occur (Chang et al., 2005), while the active sensors are equipped with internal light source and will not be affected by solar illumination variation. Second, with the same measurement height above the canopy (1.0 m), ASD yielded a smaller sampling area (with a diameter of around 0.44 m, which equals 0.15 m^2 and contains only one row of rice plant) than GreenSeeker (with a field-of-view of 0.61 × 0.015 m and walking distance of 3 m, which equals 1.8 m^2 and includes nearly three rows of rice plant). The difference in the field of view on the top of canopy led to different proportions of foliage and background for signature capture by the sensors.

4.4. *Potential applications of remote sensing based phenology estimation*

The potential use of our work is summarized in Fig. 7. The maximum, zero-crossing and minimum points of the FD of NDVI temporal profile can be determined as the dates of active tillering, middle heading and maturity, respectively. Additionally, the maximum, zero-crossing and minimum points of the FD of $CI_{red\ edge}$ temporal profile can be determined as the dates of jointing, middle booting and dough grain. Furthermore, irrigation scheduling and fertilizer management can benefit from accurate phenological dates estimation in this study. For instance, active tillering date (FD_{max} of NDVI) can be used to guide field drying, which improves root activity and optimizes canopy structure (Yang et al., 2006). Jointing date (FD_{max} of $CI_{red\ edge}$) can be useful for the panicle fertilizer decision, which has a great impact on rice grain yield and the utilization of N (Ding et al., 2003). Normally rice is harvested prior to full maturity for the entire field as the grain filling process may not be uniform. Late harvesting has negative influence on the milling quality (Moldenhauer and Slaton, 2001). Therefore, accurate detection of maturity date can guide farmers to harvest at an appropriate time.

Although additional data from more ecological stations and cultivars is needed to validate the presented results in the future work, this study could be useful for assessing crop phenology over large agricultural lands using satellite imagery given that the NDVI data could be acquired by sensors such as MODIS, AVHRR, and SPOT. The method in this study would also provide useful reference for further studies on other crops or vegetation species. The findings from this three-year consecutive experiment demonstrated the great potential of ground-based remote sensing for accurate detection of crop phenology. The future work is directed to the incorporation of NDVI and $CI_{red\ edge}$ into an operational system for continuous monitoring of crop phenology. In addition, new spectral indices with optimized wavelengths from ASD hyperspectral data will be exploited for crop phenology detection in the near future.

5. Conclusions

In this study, we used the temporal $CI_{red\ edge}$ and NDVI data smoothed with the double logistic regression function to represent intra-annual vegetation dynamics. Phenological dates were detected through extracting the maximum increase, inflection and maximum decrease points (FD_{max}, FD_{zero} and FD_{min}) of smoothed VI temporal profiles. This method was applied to $CI_{red\ edge}$ and NDVI data over a 3-year period (2013–2015) for two different cultivars with different planting densities and N rates. We found that $CI_{red\ edge}$ can be used to accurately estimate the dates of jointing, middle booting and dough grain and NDVI can be used to detect the dates of active tillering, middle heading and maturity. Between the NDVIs from passive and active sensors, $NDVI_{GS}$ yielded better performance than $NDVI_{ASD}$ for estimating the three specific phenological dates. Meanwhile, rice cultivar and N rate exhibited more impact on the estimation accuracy than growing season and planting density. The dates of active tillering, jointing and maturity could be useful for irrigation management, fertilization recommendation and harvesting determination, respectively. Our work has great potential for providing valuable support to assessing crop growth status and providing precise management strategy.

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